

3D Coordinates · Lines & Rectangles on the Wall (x, y) — English Worksheet

Topic: Lines & Rectangles on the Wall (x, y) · **Course:** 3D Coordinates · **Time:** about 45 minutes

This lesson you use a **loop** to draw a **line** of blocks: grow **x** to go across, grow **y** to go up. Four lines make a **rectangle outline** — a picture frame.

1 · Predict 🧠

Read each set of steps. Write what you think you will see.

```
set x to 0
repeat 5 times:
  place stone block at (x, 0)
  add 1 to x
```

Which way does this line grow — across or up? How long is it? Which number stays the same?


```
set y to 0
repeat 4 times:
  place gold block at (3, y)
  add 1 to y
```

This time the line is at x = 3 and y grows. Across or up? How tall is it?


```
set x to 0
repeat 6 times:
  place stone block at (x, 0)
  place stone block at (x, 4)
  add 1 to x
```

Each turn places two blocks at the same x — one at $y = 0$, one at $y = 4$. What do the blocks make — one line or two lines? How far apart, top to bottom?

2 · Spot the Bug 🐛

Each code block below is broken. Read what it should do, fix it, then explain why the original was wrong and your fix works.

Bug A — The line should grow **up** at $x = 2$, from $(2, 0)$ to $(2, 4)$. Right now it grows the wrong way.

```
set n to 0
repeat 5 times:
  place stone block at (n, 2)
  add 1 to n
```

Hint: look at which slot n is in — the x slot (across) or the y slot (up).

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The bottom edge of a frame should be **6 blocks long**, from $(0, 0)$ to $(5, 0)$. Right now it comes out one block short.

```
set x to 0
repeat 5 times:
  place gold block at (x, 0)
  add 1 to x
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This should make a line of 4 stone blocks growing up from (3, 0). Right now all the blocks land in the **same spot**.

```
set y to 0
repeat 4 times:
  place stone block at (3, y)
```

Hint: read every line of the loop — is anything growing y?

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Switch to your homework world. Build a **rectangle outline** on the wall — like an empty picture frame — using loops for the four edges (bottom, top, left, right).

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.